Unit 2 Progress Check: FRQ Part A

This question involves processing a String that represents a sequence of beads.

The String variable beadSequence represents a bead sequence. It contains a sequence of letters in which each letter represents the color of a bead.

A bead sequence is considered acceptable if it adheres to the following rules. Otherwise, it is not considered acceptable.

- Rule 1: The number of beads in the sequence is between 5 and 10, inclusive.
- Rule 2: The sequence contains at least one blue bead (represented by "B") and at least one white bead (represented by "W").
- Rule 3: The sequence starts and ends with the same color.

You will create three boolean variables to help determine whether the sequence is acceptable, and then print a message indicating the result. The following table illustrates the intended behavior for different values of beadSequence.

Value of beadSequence	Meets Rule 1	Meets Rule 2	Meets Rule 3	Printed Output
"OBWO"	false	true	true	"Not acceptable"
"BWBGOGWBB"	true	true	true	"Acceptable"
"ROYGBIVW"	true	true	false	"Not acceptable"
"BOBOBOBO"	true	false	false	"Not acceptable"
"WBWRWBWRWBWR"	false	true	false	"Not acceptable"
"BWWWBBWWWB"	true	true	true	"Acceptable"

1. Write a code segment that creates a boolean variable named lengthGood and assigns it true if beadSequence meets rule 1 and assigns it false otherwise.

2. Write a code segment that creates a boolean variable named correctColors and assigns it true if beadSequence meets rule 2 and assigns it false otherwise.

3. Write a code segment that creates a boolean variable named startEndGood and assigns it true if beadSequence meets rule 3 and assigns it false otherwise.

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4. Write a code segment that prints "Acceptable" if beadSequence adheres to all three rules and prints "Not acceptable" otherwise. You must use the variables lengthGood, correctColors, and startEndGood to receive full credit, regardless of what you wrote for parts A, B, and C.